tent_count

```r
library(ggplot2)
library(ggpubr)
library(plotrix)
library(tidyverse)
library(dplyr)
library(car)
library(lme4)
library(emmeans)
library(qqplotr)
library(here)
library(scales)
library(janitor)
library(gt)
library(ggh4x)

rm(list = ls())
#graphics.off()


getwd()
```

```
## [1] "/Users/junbc/Documents/GitHub/heating_lacerates_final/code"
```

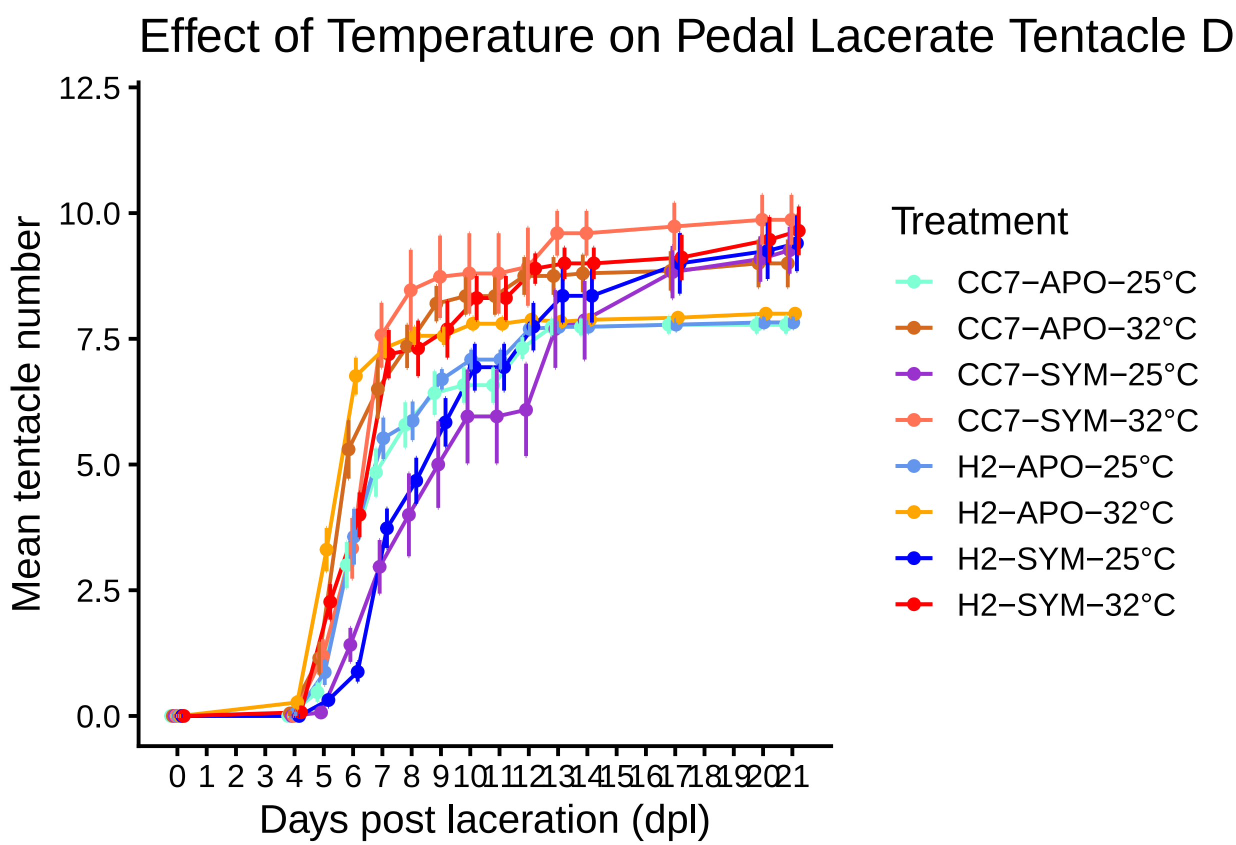
```r
setwd("~/Documents/GitHub/heating_lacerates_final")

select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate
lag <- dplyr::lag

## Experiment 2022

tentcount_2022 <- read.csv("~/Documents/GitHub/heating_lacerates_final/data/2022_tentcount.csv")

str(tentcount_2022)
```

```
## 'data.frame':    4635 obs. of  10 variables:
##  $ ID        : chr  "CC7-APO-1_B3.2" "CC7-APO-1_B3.2" "CC7-APO-1_B3.2" "CC7-APO-1_B3.2" ...
##  $ plate     : chr  "CC7-APO-1" "CC7-APO-1" "CC7-APO-1" "CC7-APO-1" ...
##  $ treatment : chr  "CC7-APO-25C" "CC7-APO-25C" "CC7-APO-25C" "CC7-APO-25C" ...
##  $ line      : chr  "CC7" "CC7" "CC7" "CC7" ...
##  $ symbiosis : chr  "Apo" "Apo" "Apo" "Apo" ...
##  $ well      : chr  "B3" "B3" "B3" "B3" ...
##  $ temp      : chr  "25C (ambient)" "25C (ambient)" "25C (ambient)" "25C (ambient)" ...
##  $ day       : int  0 4 5 6 7 8 9 10 11 12 ...
```

```
##  $ day_cat   : chr  "day_0" "day_4" "day_5" "day_6" ...
##  $ tent_count: chr  "0" "0" "0" "0" ...
```

```
tentcount_2022$tent_count <- as.numeric(tentcount_2022$tent_count)
tentcount_2022$ID <- as.factor(tentcount_2022$ID)
tentcount_2022$plate <- as.factor(tentcount_2022$plate)
tentcount_2022$well <- as.factor(tentcount_2022$well)
tentcount_2022$line <- as.factor(tentcount_2022$line)
tentcount_2022$temp <- as.factor(tentcount_2022$temp)
tentcount_2022$treatment <- as.factor(tentcount_2022$treatment)
tentcount_2022$symbiosis <- as.factor(tentcount_2022$symbiosis)
tentcount_2022$day <- as.numeric(tentcount_2022$day)
tentcount_2022$day_cat <- as.factor(tentcount_2022$day_cat)

tentcount_2022 <- tentcount_2022 %>%
  mutate(Day = as.factor(day)) %>%
  mutate(Day = dplyr::recode(Day, "0" = "00"))

tentcount_2022_cleaned <- tentcount_2022

data_means <- tentcount_2022_cleaned %>%
  group_by(treatment, day) %>%
  summarise(mean = mean(tent_count, na.rm=TRUE),
            se = std.error(tent_count, na.rm=TRUE))

All_2022_plot <- ggplot(data = data_means, aes(x = day, y = mean)) +
  theme_classic(base_size = 15) +
  geom_line(aes(color = treatment, group = treatment),
            position = position_dodge(0.5)) +
  ylab(bquote("Mean tentacle number")) +
  xlab("Days post laceration (dpl)") +
  ggtitle("Effect of Temperature on Pedal Lacerate Tentacle Development in Aiptasia") +
  ylim(0,12) +
  geom_point(aes(color = treatment),
             size = 2.5,
             shape = 20,
             position = position_dodge(0.5)) +
  scale_x_continuous(
    breaks = round(seq(min(data_means$day), max(data_means$day), by = 1), 1)
  ) +
  geom_errorbar(
    aes(color = treatment, ymin = mean - se, ymax = mean + se),
    width = 0.2,
    position = position_dodge(0.5)
  ) +
  scale_color_manual(
    values = c(
      "CC7-APO-25C" = "aquamarine",
      "CC7-APO-32C" = "chocolate",
      "CC7-SYM-25C" = "darkorchid",
      "CC7-SYM-32C" = "coral1",
      "H2-APO-25C"  = "cornflowerblue",
      "H2-APO-32C"  = "orange",
      "H2-SYM-25C"  = "blue",
```

```
      "H2-SYM-32C"   = "red"
    ),
    breaks = c(
      "CC7-APO-25C","CC7-APO-32C",
      "CC7-SYM-25C","CC7-SYM-32C",
      "H2-APO-25C","H2-APO-32C",
      "H2-SYM-25C","H2-SYM-32C"
    ),
    labels = c(
      "CC7-APO-25°C",
      "CC7-APO-32°C",
      "CC7-SYM-25°C",
      "CC7-SYM-32°C",
      "H2-APO-25°C",
      "H2-APO-32°C",
      "H2-SYM-25°C",
      "H2-SYM-32°C"
    )
  ) +
  theme(legend.text.align = 0) +
  scale_size_manual(values = c(1.2,1.2,1.2,1.2)) +
  labs(colour = "Treatment")

All_2022_plot
```



```r
## Tentacle count analysis
library(lme4)
library(car)
library(emmeans)
library(performance)
library(DHARMa)

data <- tentcount_2022_cleaned %>%
  filter(line != "CC7") %>%
  mutate(
    day_cat = factor(day_cat),
    temp = factor(temp),
    symbiosis = factor(symbiosis),
    ID = factor(ID)
  )

data <- tentcount_2022_cleaned

str(data)
```

```
## 'data.frame':    4635 obs. of  11 variables:
##  $ ID        : Factor w/ 309 levels "CC7-APO-1_A1",..: 8 8 8 8 8 8 8 8 8 8 ...
##  $ plate     : Factor w/ 24 levels "CC7-APO-1","CC7-APO-2",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ treatment : Factor w/ 8 levels "CC7-APO-25C",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ line      : Factor w/ 2 levels "CC7","H2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ symbiosis : Factor w/ 2 levels "Apo","Sym": 1 1 1 1 1 1 1 1 1 1 ...
##  $ well      : Factor w/ 12 levels "A1","A2","A3",..: 7 7 7 7 7 7 7 7 7 7 ...
##  $ temp      : Factor w/ 2 levels "25C (ambient)",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ day       : num  0 4 5 6 7 8 9 10 11 12 ...
##  $ day_cat   : Factor w/ 15 levels "day_0","day_10",..: 1 10 11 12 13 14 15 2 3 4 ...
##  $ tent_count: num  0 0 0 0 1 3 5 5 5 6 ...
##  $ Day       : Factor w/ 15 levels "00","4","5","6",..: 1 2 3 4 5 6 7 8 9 10 ...
```

```r
model_pois <- glmer(
  tent_count ~ temp * symbiosis * day_cat + (1 | ID),
  family = poisson(link = "log"),
  data = data,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)

summary(model_pois)
```

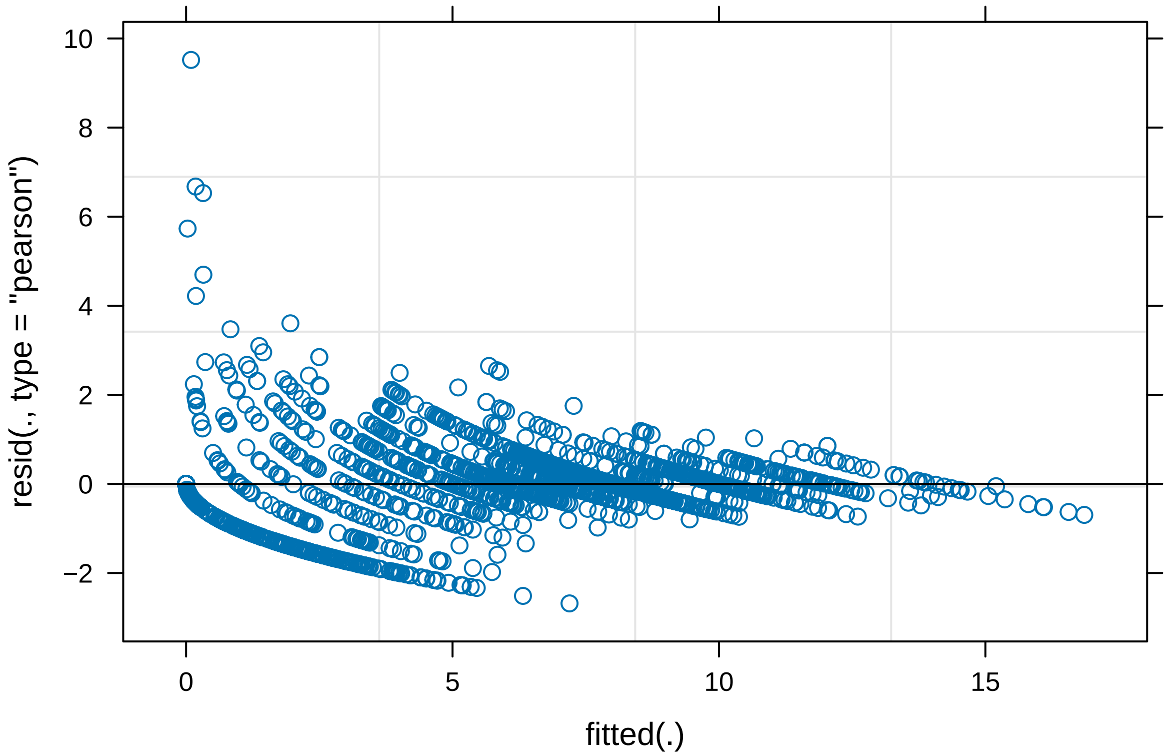
```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: poisson  ( log )
## Formula: tent_count ~ temp * symbiosis * day_cat + (1 | ID)
##    Data: data
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
```

```
##   10635.5   11001.2   -5256.7   10513.5      2904
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.6825 -0.2409  0.0000  0.2174  9.5187
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  ID     (Intercept) 0.07849  0.2802
## Number of obs: 2965, groups:  ID, 309
##
## Fixed effects:
##                                                 Estimate Std. Error z value
## (Intercept)                                   -2.031e+01  3.512e+03  -0.006
## temp32C (heat stress)                          8.221e-03  5.004e+03   0.000
## symbiosisSym                                   1.121e-02  4.320e+03   0.000
## day_catday_10                                  2.222e+01  3.512e+03   0.006
## day_catday_11                                  2.222e+01  3.512e+03   0.006
## day_catday_12                                  2.231e+01  3.512e+03   0.006
## day_catday_13                                  2.234e+01  3.512e+03   0.006
## day_catday_14                                  2.234e+01  3.512e+03   0.006
## day_catday_17                                  2.234e+01  3.512e+03   0.006
## day_catday_20                                  2.235e+01  3.512e+03   0.006
## day_catday_21                                  2.235e+01  3.512e+03   0.006
## day_catday_4                                   1.656e+01  3.512e+03   0.005
## day_catday_5                                   1.992e+01  3.512e+03   0.006
## day_catday_6                                   2.149e+01  3.512e+03   0.006
## day_catday_7                                   2.195e+01  3.512e+03   0.006
## day_catday_8                                   2.206e+01  3.512e+03   0.006
## day_catday_9                                   2.218e+01  3.512e+03   0.006
## temp32C (heat stress):symbiosisSym            -2.764e-03  6.112e+03   0.000
## temp32C (heat stress):day_catday_10            1.501e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_11            1.501e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_12            8.454e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_13            5.376e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_14            5.914e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_17            6.353e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_20            7.365e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_21            7.365e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_4             1.981e+00  5.004e+03   0.000
## temp32C (heat stress):day_catday_5             1.225e+00  5.004e+03   0.000
## temp32C (heat stress):day_catday_6             6.037e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_7             2.785e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_8             2.372e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_9             1.675e-01  5.004e+03   0.000
## symbiosisSym:day_catday_10                    -1.017e-01  4.320e+03   0.000
## symbiosisSym:day_catday_11                    -1.017e-01  4.320e+03   0.000
## symbiosisSym:day_catday_12                    -1.179e-01  4.320e+03   0.000
## symbiosisSym:day_catday_13                    -8.517e-03  4.320e+03   0.000
## symbiosisSym:day_catday_14                     6.218e-04  4.320e+03   0.000
## symbiosisSym:day_catday_17                     1.043e-01  4.320e+03   0.000
## symbiosisSym:day_catday_20                     1.295e-01  4.320e+03   0.000
## symbiosisSym:day_catday_21                     1.471e-01  4.320e+03   0.000
## symbiosisSym:day_catday_4                     -1.655e+01  5.236e+03  -0.003
```

```
## symbiosisSym:day_catday_5                        -1.233e+00  4.320e+03   0.000
## symbiosisSym:day_catday_6                        -1.164e+00  4.320e+03   0.000
## symbiosisSym:day_catday_7                        -4.862e-01  4.320e+03   0.000
## symbiosisSym:day_catday_8                        -3.355e-01  4.320e+03   0.000
## symbiosisSym:day_catday_9                        -2.324e-01  4.320e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_10  1.400e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_11  1.400e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_12  1.787e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_13  1.046e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_14  9.008e-02  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_17 -1.014e-02  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_20 -2.234e-02  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_21 -3.029e-02  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_4   1.524e+01  6.791e+03   0.002
## temp32C (heat stress):symbiosisSym:day_catday_5   9.785e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_6   6.491e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_7   5.043e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_8   3.528e-01  6.112e+03   0.000
## temp32C (heat stress):symbiosisSym:day_catday_9   2.436e-01  6.112e+03   0.000
##                                                  Pr(>|z|)
## (Intercept)                                         0.995
## temp32C (heat stress)                               1.000
## symbiosisSym                                        1.000
## day_catday_10                                       0.995
## day_catday_11                                       0.995
## day_catday_12                                       0.995
## day_catday_13                                       0.995
## day_catday_14                                       0.995
## day_catday_17                                       0.995
## day_catday_20                                       0.995
## day_catday_21                                       0.995
## day_catday_4                                        0.996
## day_catday_5                                        0.995
## day_catday_6                                        0.995
## day_catday_7                                        0.995
## day_catday_8                                        0.995
## day_catday_9                                        0.995
## temp32C (heat stress):symbiosisSym                  1.000
## temp32C (heat stress):day_catday_10                 1.000
## temp32C (heat stress):day_catday_11                 1.000
## temp32C (heat stress):day_catday_12                 1.000
## temp32C (heat stress):day_catday_13                 1.000
## temp32C (heat stress):day_catday_14                 1.000
## temp32C (heat stress):day_catday_17                 1.000
## temp32C (heat stress):day_catday_20                 1.000
## temp32C (heat stress):day_catday_21                 1.000
## temp32C (heat stress):day_catday_4                  1.000
## temp32C (heat stress):day_catday_5                  1.000
## temp32C (heat stress):day_catday_6                  1.000
## temp32C (heat stress):day_catday_7                  1.000
## temp32C (heat stress):day_catday_8                  1.000
## temp32C (heat stress):day_catday_9                  1.000
## symbiosisSym:day_catday_10                          1.000
## symbiosisSym:day_catday_11                          1.000
```

```
## symbiosisSym:day_catday_12                          1.000
## symbiosisSym:day_catday_13                          1.000
## symbiosisSym:day_catday_14                          1.000
## symbiosisSym:day_catday_17                          1.000
## symbiosisSym:day_catday_20                          1.000
## symbiosisSym:day_catday_21                          1.000
## symbiosisSym:day_catday_4                           0.997
## symbiosisSym:day_catday_5                           1.000
## symbiosisSym:day_catday_6                           1.000
## symbiosisSym:day_catday_7                           1.000
## symbiosisSym:day_catday_8                           1.000
## symbiosisSym:day_catday_9                           1.000
## temp32C (heat stress):symbiosisSym:day_catday_10    1.000
## temp32C (heat stress):symbiosisSym:day_catday_11    1.000
## temp32C (heat stress):symbiosisSym:day_catday_12    1.000
## temp32C (heat stress):symbiosisSym:day_catday_13    1.000
## temp32C (heat stress):symbiosisSym:day_catday_14    1.000
## temp32C (heat stress):symbiosisSym:day_catday_17    1.000
## temp32C (heat stress):symbiosisSym:day_catday_20    1.000
## temp32C (heat stress):symbiosisSym:day_catday_21    1.000
## temp32C (heat stress):symbiosisSym:day_catday_4     0.998
## temp32C (heat stress):symbiosisSym:day_catday_5     1.000
## temp32C (heat stress):symbiosisSym:day_catday_6     1.000
## temp32C (heat stress):symbiosisSym:day_catday_7     1.000
## temp32C (heat stress):symbiosisSym:day_catday_8     1.000
## temp32C (heat stress):symbiosisSym:day_catday_9     1.000

## optimizer (bobyqa) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate  Hessian with 4 negative eigenvalues
```

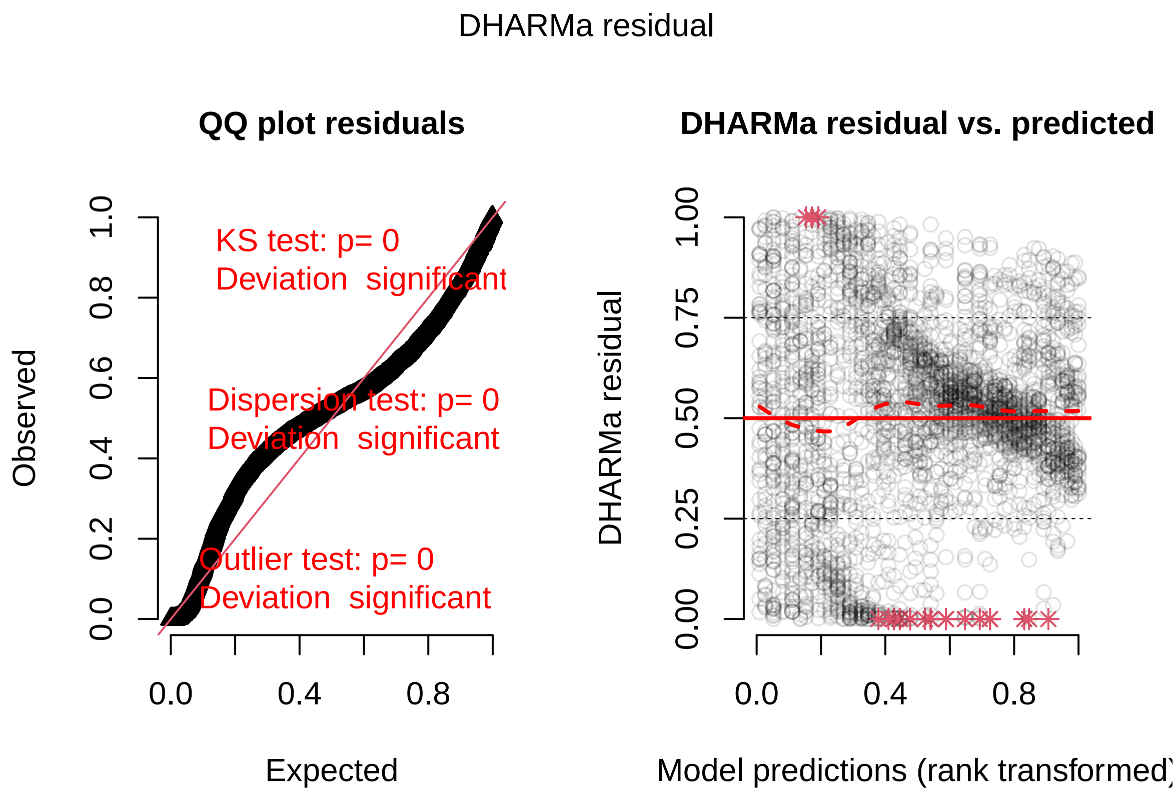
```r
plot(model_pois)
```




```r
check_overdispersion(model_pois)
```

```
## # Overdispersion test
##
##        dispersion ratio =    0.529
##   Pearson's Chi-Squared = 1537.053
##                 p-value =        1
```

```r
sim_res <- simulateResiduals(model_pois)
plot(sim_res)
```




```r
anova_tentacle <- Anova(model_pois, type = "II")

anova_tentacle
```

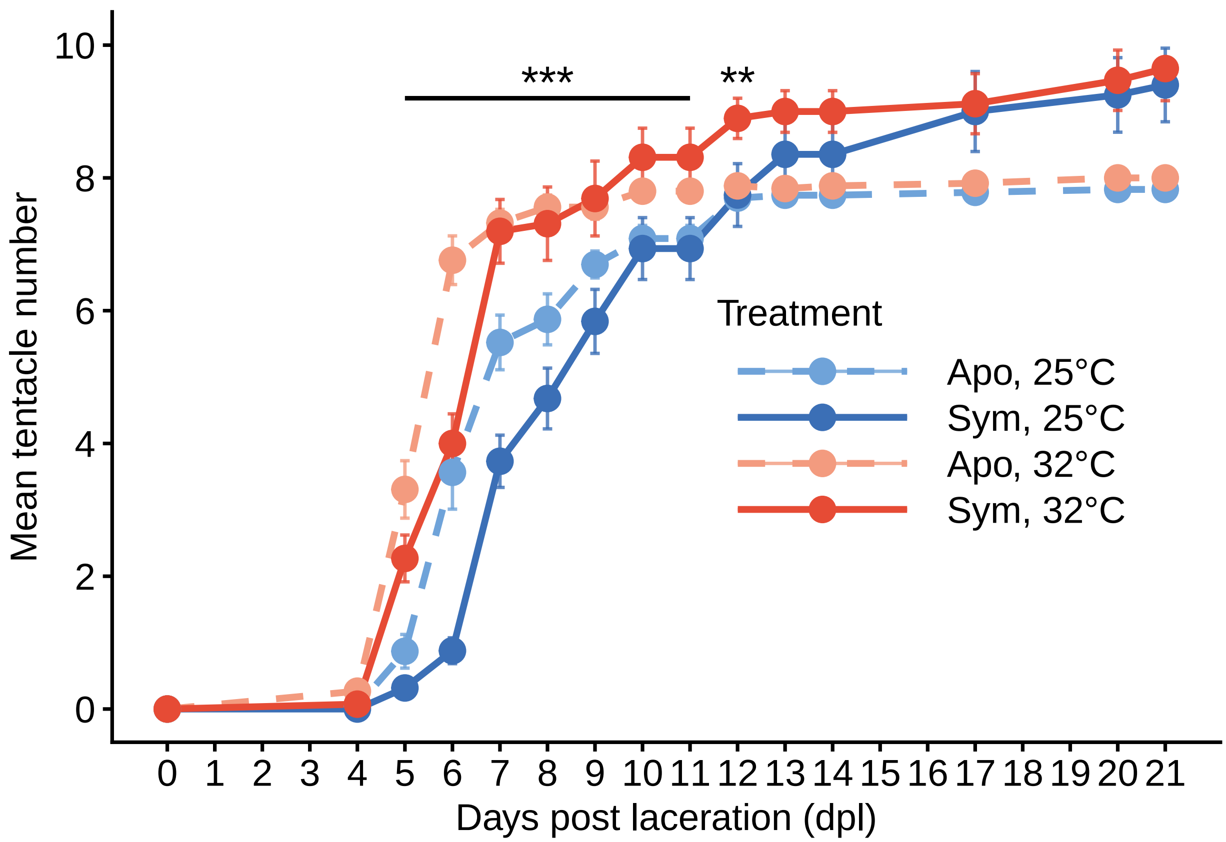
```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: tent_count
##                           Chisq Df Pr(>Chisq)
## temp                    287.089 27  < 2.2e-16 ***
## symbiosis               173.768 27  < 2.2e-16 ***
## day_cat                1801.773 52  < 2.2e-16 ***
## temp:symbiosis           38.127 13  0.0002753 ***
## temp:day_cat            236.544 26  < 2.2e-16 ***
## symbiosis:day_cat       170.146 26  < 2.2e-16 ***
## temp:symbiosis:day_cat   33.365 14  0.0025514 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova_tentacle_df <- as.data.frame(anova_tentacle) %>%
  tibble::rownames_to_column("Factor")

write.csv(
  anova_tentacle_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
```

```
    "TableS1_anova_tentacle_number.csv"
  ),
  row.names = FALSE
)

emm_temp <- emmeans(model_pois, ~ temp | day_cat, type = "response")
tukey_temp <- pairs(emm_temp, adjust = "tukey")

tukey_temp_df <- as.data.frame(tukey_temp)

write.csv(
  tukey_temp_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS2_tukey_temperature_tentacle_number.csv"
  ),
  row.names = FALSE
)

emm_symbiosis <- emmeans(model_pois, ~ symbiosis | day_cat, type = "response")
tukey_sym <- pairs(emm_symbiosis, adjust = "tukey")

tukey_sym_df <- as.data.frame(tukey_sym)


fig1_dat_raw <- tentcount_2022_cleaned %>%
  filter(line != "CC7") %>%
  mutate(
    day_cat = factor(day_cat),
    temp = factor(temp),
    symbiosis = factor(symbiosis),
    ID = factor(ID)
  )

fig1fuller_data <- fig1_dat_raw %>%
  group_by(temp, symbiosis, day) %>%
  summarise(
    mean = mean(tent_count, na.rm = TRUE),
    se = std.error(tent_count, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    treatment = case_when(
      symbiosis == "Apo" & temp == "25C (ambient)" ~ "Apo, 25°C",
      symbiosis == "Sym" & temp == "25C (ambient)" ~ "Sym, 25°C",
      symbiosis == "Apo" & temp == "32C (heat stress)" ~ "Apo, 32°C",
      symbiosis == "Sym" & temp == "32C (heat stress)" ~ "Sym, 32°C"
    ),
    treatment = factor(
      treatment,
      levels = c("Apo, 25°C", "Sym, 25°C", "Apo, 32°C", "Sym, 32°C")
    )
  )
```

```r
my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

fig1_fuller <- ggplot(
  fig1fuller_data,
  aes(
    x = day,
    y = mean,
    color = treatment,
    linetype = treatment,
    group = treatment
  )
) +
  my_theme +
  geom_line(linewidth = 1.2) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6,
    alpha = 0.8,
    linetype = "solid"
  ) +
  geom_point(size = 4) +

annotate("segment", x = 5, xend = 11, y = 9.2, yend = 9.2, linewidth = 0.8) +
  annotate("text", x = 8, y = 9.45, label = "***", size = 6) +
  annotate("text", x = 12, y = 9.45, label = "**", size = 6) +

  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +
  scale_y_continuous(
    breaks = seq(0, 10, 2),
    limits = c(0, 10)
  ) +
  scale_x_continuous(
    breaks = seq(min(fig1fuller_data$day), max(fig1fuller_data$day), 1)
  ) +
  scale_color_manual(
    values = c(
      "Apo, 25°C" = "#6FA3D9",
      "Sym, 25°C" = "#3B6FB6",
      "Apo, 32°C" = "#F39B7F",
      "Sym, 32°C" = "#E64B35"
    )
  ) +
  scale_linetype_manual(
    values = c(
```

```
      "Apo, 25°C" = "dashed",
      "Sym, 25°C" = "solid",
      "Apo, 32°C" = "dashed",
      "Sym, 32°C" = "solid"
    )
  ) +
  labs(
    color = "Treatment",
    linetype = "Treatment"
  ) +
  theme(
    legend.position = c(0.73, 0.45),
    legend.justification = c("center", "center"),
    legend.key.width = unit(2.8, "cm")
  )

fig1_fuller
```



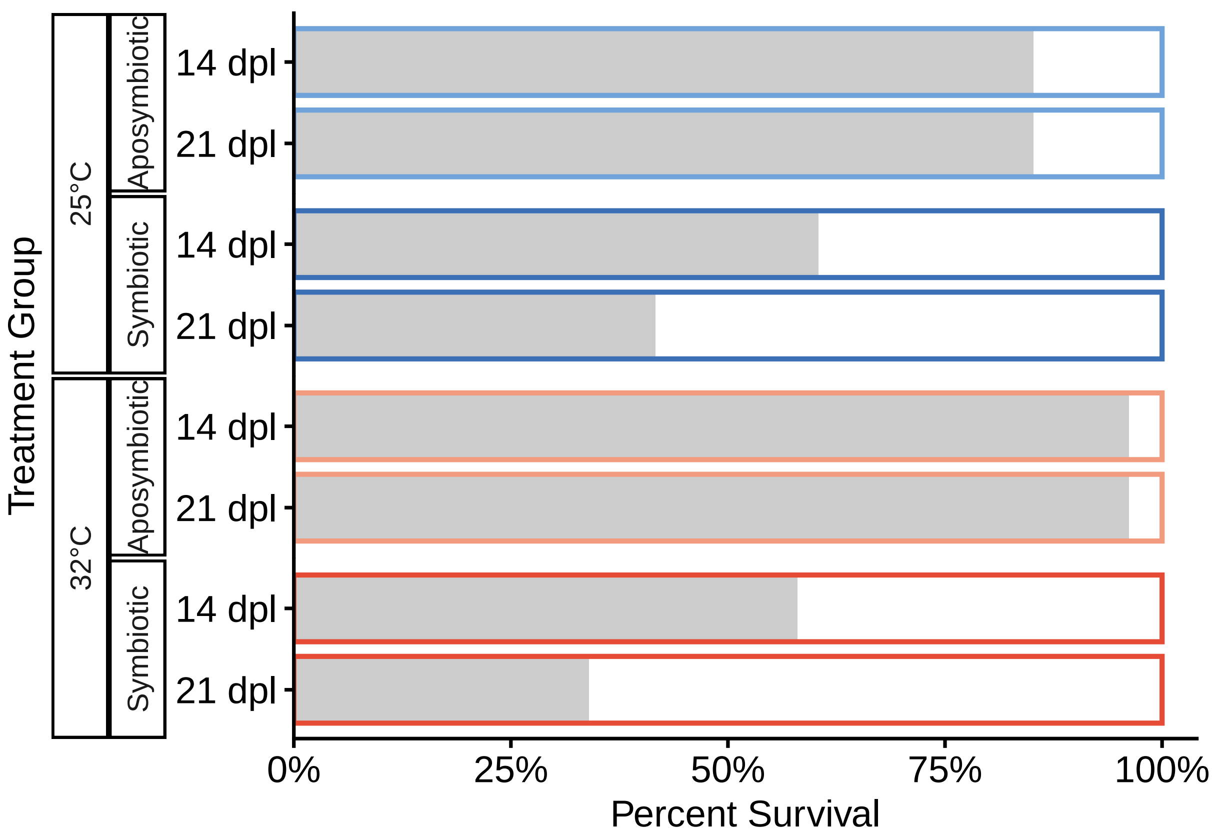

```
ggsave(
  filename = "Fig2.png",
  plot = fig1_fuller,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
```

```r
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "Fig2.pdf",
  plot = fig1_fuller,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "pdf",
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)


## 2022 mortality figure

df_mortality_2022 <- tentcount_2022_cleaned %>%
  filter(
    line == "H2",
    day_cat %in% c("day_14", "day_21")
  ) %>%
  mutate(
    Mortality = ifelse(is.na(tent_count) | tent_count == 0, "Dead", "Alive"),
    dead = ifelse(Mortality == "Dead", 1, 0),
    Mortality = factor(Mortality, levels = c("Alive", "Dead")),
    day_cat = factor(day_cat, levels = c("day_14", "day_21"))
  )

my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

mortality_2022_plot_df_flipped <- df_mortality_2022 %>%
  mutate(
    sym_state = case_when(
      grepl("APO", treatment) ~ "Aposymbiotic",
      grepl("SYM", treatment) ~ "Symbiotic"
    ),
    temp_label = case_when(
      grepl("25", treatment) ~ "25°C",
      grepl("32", treatment) ~ "32°C"
    ),
    day_label = case_when(
      day_cat == "day_14" ~ "14 dpl",
      day_cat == "day_21" ~ "21 dpl"
```

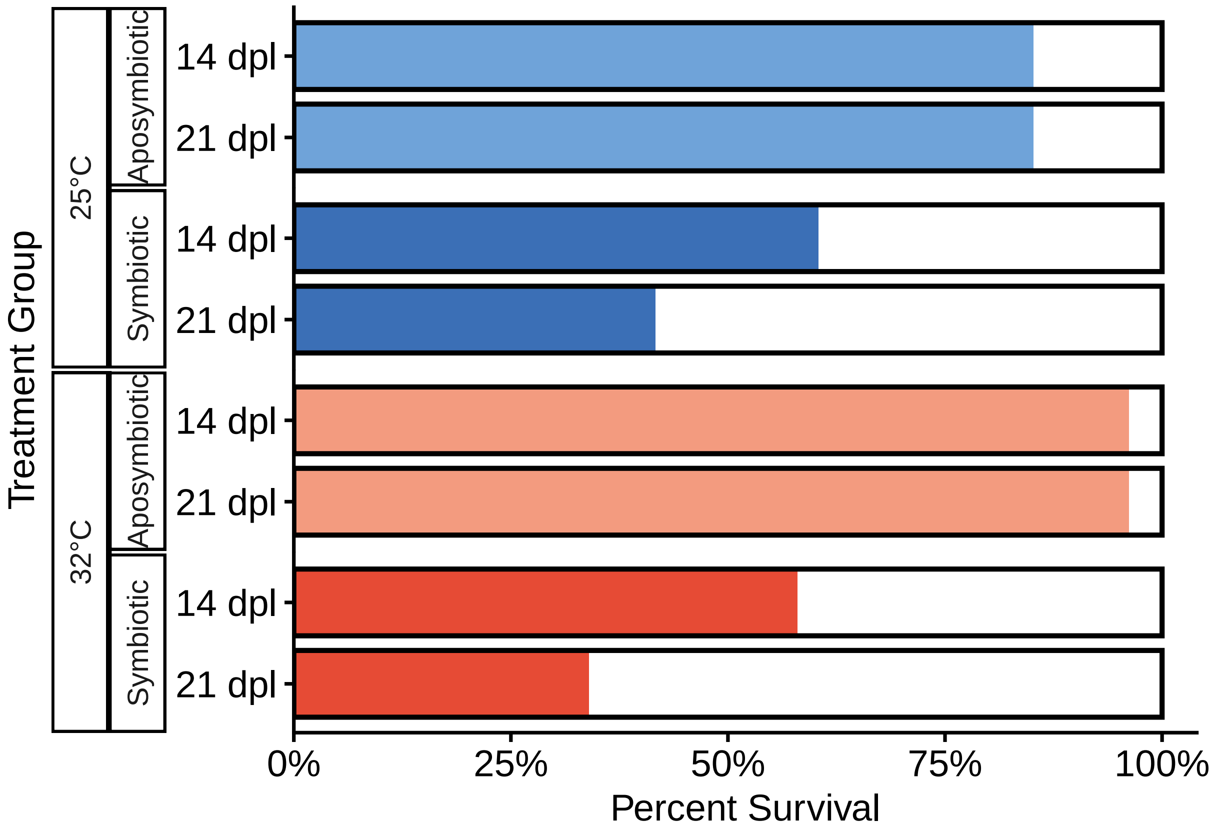
```
    )
  ) %>%
  group_by(sym_state, temp_label, day_label) %>%
  summarise(
    mortality_prop = mean(dead, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    survival_prop = 1 - mortality_prop,
    temp_label = factor(temp_label, levels = c("25°C", "32°C")),
    sym_state = factor(sym_state, levels = c("Aposymbiotic", "Symbiotic")),
    outline_group = case_when(
      sym_state == "Aposymbiotic" & temp_label == "25°C" ~ "apo_25",
      sym_state == "Symbiotic"    & temp_label == "25°C" ~ "sym_25",
      sym_state == "Aposymbiotic" & temp_label == "32°C" ~ "apo_32",
      sym_state == "Symbiotic"    & temp_label == "32°C" ~ "sym_32"
    ),
    group_order = case_when(
      sym_state == "Aposymbiotic" & day_label == "14 dpl" ~ 1,
      sym_state == "Aposymbiotic" & day_label == "21 dpl" ~ 2,
      sym_state == "Symbiotic"    & day_label == "14 dpl" ~ 3,
      sym_state == "Symbiotic"    & day_label == "21 dpl" ~ 4
    ),
    axis_text = case_when(
      day_label == "14 dpl" ~ "14 dpl",
      day_label == "21 dpl" ~ "21 dpl"
    )
  ) %>%
  arrange(temp_label, group_order) %>%
  group_by(temp_label, sym_state) %>%
  mutate(
    bar_label = factor(axis_text, levels = rev(unique(axis_text)))
  ) %>%
  ungroup()

survival_fill_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

outline_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

p_survival_2022_base <- ggplot(
  mortality_2022_plot_df_flipped,
  aes(x = bar_label)
) +
```

```r
  coord_flip() +
  ggh4x::facet_nested(
    rows = vars(temp_label, sym_state),
    scales = "free_y",
    space = "free_y",
    switch = "y"
  ) +
  scale_y_continuous(
    labels = scales::percent,
    limits = c(0, 1),
    expand = expansion(mult = c(0, 0.04))
  ) +
  labs(
    x = "Treatment Group",
    y = "Percent Survival"
  ) +
  my_theme +
  theme(
    panel.grid = element_blank(),
    strip.placement = "outside",
    strip.text.y.right = element_text(size = 14, face = "bold"),
    strip.background.y = element_rect(
      fill = "white",
      color = "black",
      linewidth = 1
    ),
    ggh4x.facet.nestline = element_blank(),
    panel.spacing.y = unit(0.08, "lines"),
    axis.text.y = element_text(size = 14, lineheight = 0.9),
    axis.text.x = element_text(size = 14),
    plot.margin = margin(8, 14, 8, 8)
  )

p_survival_2022_flipped_outline <- p_survival_2022_base +
  geom_col(
    aes(y = survival_prop),
    fill = "grey80",
    width = 0.82
  ) +
  geom_col(
    aes(y = 1, color = outline_group),
    fill = NA,
    linewidth = 0.9,
    width = 0.82
  ) +
  scale_color_manual(
    values = outline_colors,
    guide = "none"
  )

p_survival_2022_flipped_colorbar <- p_survival_2022_base +
  geom_col(
    aes(y = survival_prop, fill = outline_group),
```

```
    width = 0.82,
    color = NA
  ) +
  geom_col(
    aes(y = 1),
    width = 0.82,
    fill = NA,
    color = "black",
    linewidth = 0.9
  ) +
  scale_fill_manual(
    values = survival_fill_colors,
    guide = "none"
  )

p_survival_2022_flipped_outline
```




```
p_survival_2022_flipped_colorbar
```




```r
ggsave(
  filename = "Fig3_2022.png",
  plot = p_survival_2022_flipped_colorbar,
  path = "figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "Fig3_2022.pdf",
  plot = p_survival_2022_flipped_colorbar,
  path = "figs",
  device = pdf,
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)

## 2022 survival stats

survival_stats_2022 <- df_mortality_2022 %>%
```

```r
  mutate(
    alive = 1 - dead,
    sym_state = case_when(
      grepl("APO", treatment) ~ "Aposymbiotic",
      grepl("SYM", treatment) ~ "Symbiotic"
    ),
    temp_label = case_when(
      grepl("25", treatment) ~ "25°C",
      grepl("32", treatment) ~ "32°C"
    )
  ) %>%
  filter(
    !is.na(sym_state),
    !is.na(temp_label)
  ) %>%
  mutate(
    sym_state = factor(
      sym_state,
      levels = c("Aposymbiotic", "Symbiotic")
    ),
    temp_label = factor(
      temp_label,
      levels = c("25°C", "32°C")
    )
  )

survival_glm_2022 <- glm(
  alive ~ temp_label * sym_state,
  data = survival_stats_2022,
  family = binomial
)

summary(survival_glm_2022)
```

```
##
## Call:
## glm(formula = alive ~ temp_label * sym_state, family = binomial,
##     data = survival_stats_2022)
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                      1.7492     0.3831   4.566 4.96e-06 ***
## temp_label32°C                   1.4697     0.8163   1.800   0.0718 .
## sym_stateSymbiotic              -1.7075     0.4341  -3.934 8.37e-05 ***
## temp_label32°C:sym_stateSymbiotic  -1.6717     0.8651  -1.932   0.0533 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 397.21  on 301  degrees of freedom
## Residual deviance: 333.29  on 298  degrees of freedom
## AIC: 341.29
```

```
##
## Number of Fisher Scoring iterations: 5
```

```
table_survival_2022 <- summary(survival_glm_2022)$coefficients %>%
  as.data.frame() %>%
  tibble::rownames_to_column("Term")

colnames(table_survival_2022) <- c(
  "Term",
  "Estimate",
  "SE",
  "z",
  "p"
)

write.csv(
  table_survival_2022,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS3_survival_glm_2022.csv"
  ),
  row.names = FALSE
)

emm_surv_temp_2022 <- emmeans(
  survival_glm_2022,
  ~ temp_label | sym_state
)

emm_surv_sym_2022 <- emmeans(
  survival_glm_2022,
  ~ sym_state | temp_label
)

emm_surv_temp_2022
```

```
## sym_state = Aposymbiotic:
##  temp_label  emmean    SE  df asymp.LCL asymp.UCL
##  25°C        1.7492 0.383 Inf     0.998     2.500
##  32°C        3.2189 0.721 Inf     1.806     4.632
##
## sym_state = Symbiotic:
##  temp_label  emmean    SE  df asymp.LCL asymp.UCL
##  25°C        0.0417 0.204 Inf    -0.358     0.442
##  32°C       -0.1603 0.201 Inf    -0.554     0.233
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
emm_surv_sym_2022
```

```
## temp_label = 25°C:
##  sym_state     emmean    SE  df asymp.LCL asymp.UCL
```

```
##  Aposymbiotic  1.7492 0.383 Inf     0.998     2.500
##  Symbiotic     0.0417 0.204 Inf    -0.358     0.442
##
## temp_label = 32°C:
##  sym_state     emmean    SE  df asymp.LCL asymp.UCL
##  Aposymbiotic  3.2189 0.721 Inf     1.806     4.632
##  Symbiotic    -0.1603 0.201 Inf    -0.554     0.233
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
## Supplementary heat ramp

library(ggplot2)
library(plotrix)
library(dplyr)
library(janitor)
library(here)
library(ggh4x)

#graphics.off()

setwd("~/Documents/GitHub/heating_lacerates_final")

supp_heat_ramp_long_collapsed <- read.csv("~/Documents/GitHub/heating_lacerates_final/data/heat ramp.cs

str(supp_heat_ramp_long_collapsed)
```

```
## 'data.frame':    3925 obs. of  10 variables:
##  $ ID        : chr  "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" ...
##  $ plate     : chr  "H2-Sym-1" "H2-Sym-1" "H2-Sym-1" "H2-Sym-1" ...
##  $ treatment : chr  "H2-SYM-25C (a)" "H2-SYM-25C (a)" "H2-SYM-25C (a)" "H2-SYM-25C (a)" ...
##  $ line      : chr  "H2" "H2" "H2" "H2" ...
##  $ symbiosis : chr  "Sym" "Sym" "Sym" "Sym" ...
##  $ well      : chr  "A1" "A1" "A1" "A1" ...
##  $ temp      : chr  "25C (ambient)" "25C (ambient)" "25C (ambient)" "25C (ambient)" ...
##  $ day       : int  0 1 2 3 4 5 6 7 8 9 ...
##  $ tent_count: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ day_cat   : chr  "day_0" "day_1" "day_2" "day_3" ...
```

```
supp_heat_ramp_long_collapsed$tent_count <- as.numeric(supp_heat_ramp_long_collapsed$tent_count)
supp_heat_ramp_long_collapsed$ID <- as.factor(supp_heat_ramp_long_collapsed$ID)
supp_heat_ramp_long_collapsed$plate <- as.factor(supp_heat_ramp_long_collapsed$plate)
supp_heat_ramp_long_collapsed$well <- as.factor(supp_heat_ramp_long_collapsed$well)
supp_heat_ramp_long_collapsed$line <- as.factor(supp_heat_ramp_long_collapsed$line)
supp_heat_ramp_long_collapsed$temp <- as.factor(supp_heat_ramp_long_collapsed$temp)
supp_heat_ramp_long_collapsed$treatment <- as.character(supp_heat_ramp_long_collapsed$treatment)
supp_heat_ramp_long_collapsed$symbiosis <- as.factor(supp_heat_ramp_long_collapsed$symbiosis)
supp_heat_ramp_long_collapsed$day <- as.numeric(supp_heat_ramp_long_collapsed$day)
supp_heat_ramp_long_collapsed$day_cat <- as.factor(supp_heat_ramp_long_collapsed$day_cat)

supp_heat_ramp_data <- supp_heat_ramp_long_collapsed %>%
  filter(
```

```r
    treatment %in% c(
      "H2-SYM-25C",
      "H2-SYM-25C (a)",
      "H2-SYM-25C (b)",
      "H2-SYM-25C (c)",
      "H2-SYM-32C",
      "H2-SYM-33.5C",
      "H2-SYM-35C"
    ),
    day <= 14
  ) %>%
  mutate(
    temp = gsub("H2-SYM-", "", treatment),
    temp = gsub(" \\(.*\\)", "", temp),
    temp = factor(
      temp,
      levels = c("25C", "32C", "33.5C", "35C")
    )
  )

supp_heat_ramp_counts_rows <- supp_heat_ramp_data %>%
  count(temp, name = "n_rows")

print(supp_heat_ramp_counts_rows)
```

```
##    temp n_rows
## 1   25C   1800
## 2   32C    750
## 3 33.5C    540
## 4   35C    540
```

```r
supp_heat_ramp_counts_ids <- supp_heat_ramp_data %>%
  distinct(ID, temp) %>%
  count(temp, name = "n_ids")

print(supp_heat_ramp_counts_ids)
```

```
##    temp n_ids
## 1   25C   120
## 2   32C    50
## 3 33.5C    36
## 4   35C    36
```

```r
supp_heat_ramp_fig_data <- supp_heat_ramp_data %>%
  group_by(temp, day) %>%
  summarise(
    mean = mean(tent_count, na.rm = TRUE),
    se = std.error(tent_count, na.rm = TRUE),
    .groups = "drop"
  )

print(supp_heat_ramp_fig_data, n = Inf)
```

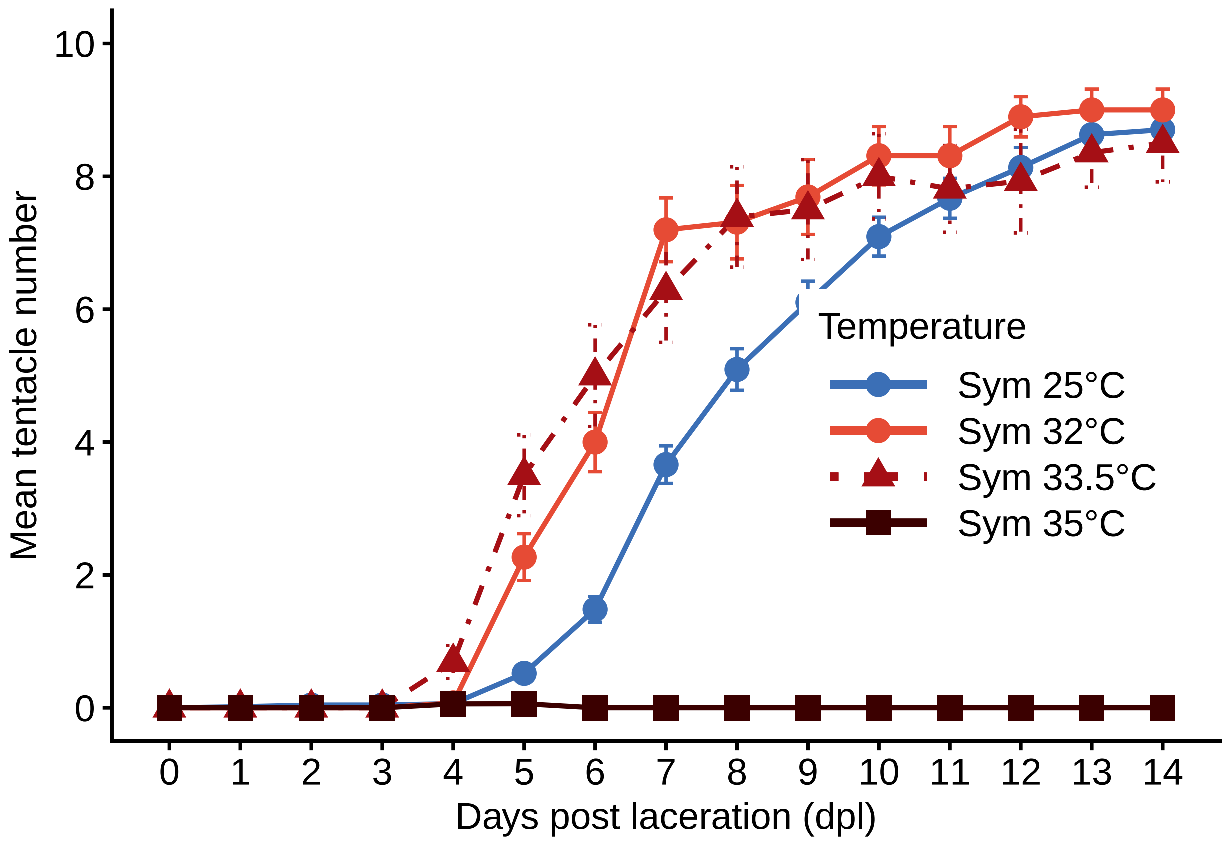
```
## # A tibble: 60 x 4
##    temp    day   mean     se
##    <fct> <dbl>  <dbl>  <dbl>
##  1 25C       0 0      0
##  2 25C       1 0.0167 0.0167
##  3 25C       2 0.0420 0.0302
##  4 25C       3 0.0424 0.0304
##  5 25C       4 0.0631 0.0410
##  6 25C       5 0.518  0.115
##  7 25C       6 1.48   0.193
##  8 25C       7 3.66   0.282
##  9 25C       8 5.09   0.313
## 10 25C       9 6.10   0.319
## 11 25C      10 7.09   0.292
## 12 25C      11 7.67   0.301
## 13 25C      12 8.13   0.300
## 14 25C      13 8.63   0.291
## 15 25C      14 8.70   0.290
## 16 32C       0 0      0
## 17 32C       1 0      0
## 18 32C       2 0      0
## 19 32C       3 0      0
## 20 32C       4 0.0732 0.0732
## 21 32C       5 2.27   0.353
## 22 32C       6 4      0.446
## 23 32C       7 7.20   0.480
## 24 32C       8 7.31   0.553
## 25 32C       9 7.69   0.564
## 26 32C      10 8.31   0.439
## 27 32C      11 8.31   0.439
## 28 32C      12 8.90   0.303
## 29 32C      13 9      0.314
## 30 32C      14 9      0.314
## 31 33.5C     0 0      0
## 32 33.5C     1 0      0
## 33 33.5C     2 0      0
## 34 33.5C     3 0      0
## 35 33.5C     4 0.690  0.249
## 36 33.5C     5 3.5    0.608
## 37 33.5C     6 5      0.765
## 38 33.5C     7 6.29   0.784
## 39 33.5C     8 7.39   0.755
## 40 33.5C     9 7.5    0.751
## 41 33.5C    10 8      0.642
## 42 33.5C    11 7.81   0.653
## 43 33.5C    12 7.93   0.781
## 44 33.5C    13 8.36   0.520
## 45 33.5C    14 8.5    0.584
## 46 35C       0 0      0
## 47 35C       1 0      0
## 48 35C       2 0      0
## 49 35C       3 0      0
## 50 35C       4 0.0588 0.0588
## 51 35C       5 0.0606 0.0606
```

```
## 52 35C       6 0      0
## 53 35C       7 0      0
## 54 35C       8 0      0
## 55 35C       9 0      0
## 56 35C      10 0      0
## 57 35C      11 0      0
## 58 35C      12 0      0
## 59 35C      13 0      0
## 60 35C      14 0      0
```

```
my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

supp_heat_ramp_fig <- ggplot(
  supp_heat_ramp_fig_data,
  aes(
    x = day,
    y = mean,
    color = temp,
    linetype = temp,
    shape = temp,
    group = temp
  )
) +
  my_theme +
  geom_line(linewidth = 0.9) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6
  ) +
  geom_point(size = 4) +
  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +
  scale_y_continuous(
    breaks = seq(0, 15, 2),
    limits = c(0, 10)
  ) +
  scale_x_continuous(
    breaks = seq(min(supp_heat_ramp_fig_data$day), max(supp_heat_ramp_fig_data$day), 1)
  ) +

  scale_color_manual(
    values = c(
      "25C"   = "#3B6FB6",
      "32C"   = "#E64B35",
      "33.5C" = "#A50F15",
      "35C"   = "#3B0000"
```

```
  ),
  labels = c(
    "25C"   = "Sym 25°C",
    "32C"   = "Sym 32°C",
    "33.5C" = "Sym 33.5°C",
    "35C"   = "Sym 35°C"
  )
) +
scale_linetype_manual(
  values = c(
    "25C"   = "solid",
    "32C"   = "solid",
    "33.5C" = "dotdash",
    "35C"   = "solid"
  ),
  labels = c(
    "25C"   = "Sym 25°C",
    "32C"   = "Sym 32°C",
    "33.5C" = "Sym 33.5°C",
    "35C"   = "Sym 35°C"
  )
) +
scale_shape_manual(
  values = c(
    "25C"   = 16,
    "32C"   = 16,
    "33.5C" = 17,
    "35C"   = 15
  ),
  labels = c(
    "25C"   = "Sym 25°C",
    "32C"   = "Sym 32°C",
    "33.5C" = "Sym 33.5°C",
    "35C"   = "Sym 35°C"
  )
) +

labs(
  color = "Temperature",
  linetype = "Temperature",
  shape = "Temperature"
) +
guides(
  color = guide_legend(
    override.aes = list(
      linewidth = 1.5,
      linetype = c("solid", "solid", "dotdash", "solid"),
      shape = c(16, 16, 17, 15)
    )
  ),
  linetype = "none",
  shape = "none"
) +
```

```r
  theme(
    legend.position = c(0.79, 0.43),
    legend.justification = c("center", "center"),
    legend.key.width = unit(1.6, "cm")
  )

supp_heat_ramp_fig
```




```r
ggsave(
  filename = "FigS1_heat_ramp.png",
  plot = supp_heat_ramp_fig,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "FigS1_heat_ramp.pdf",
  plot = supp_heat_ramp_fig,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "pdf",
```

```
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)


treatment_cols <- c(
  "apo_25"  = "#6FA3D9",
  "inoc_25" = "#3B88C3",
  "sym_25"  = "#3B6FB6",
  "apo_32"  = "#F39B7F",
  "inoc_32" = "#D95F02",
  "sym_32"  = "#E64B35"
)
```